

Highlights

Manuscript: BowTie Risk Analysis of the Valve Train in a High-Performance VW AP 1.8 Cylinder Head Operating at 10,000 rpm

Target journal: *Reliability Engineering & System Safety*

- Quantitative-coupled BowTie for high-rpm valve-train risk
- Top event: loss of valve–cam follower contact
- Outer-spring coil-bind margin 0.78 mm vs project 1.5 mm
- Inner-spring Goodman FoS 0.68 fails in 100% of MC trials
- Monte Carlo refinement under ASTM A877/A877M tolerances